Test Case Document

VendorSetu

Street Vendor Formalization & Micro-Credit Platform

Version 1.1 · Based on SRS v1.0 and Test Case Template (Doc T_01)

Prepared by: Nancy Yadav, <Member 2>, <Member 3>, <Member 4>

B.Tech CSE, Batch 2024 - CSE3005 Software Engineering - BML Munjal University

Submitted to: <Faculty name>

Test execution date: 04-11-2026

1. Test Plan Summary

Item	Details
Objective	Verify that the four implemented modules of VendorSetu meet their functional requirements (SRS Section 4.1-4.6) and reject invalid input with the correct message.
Modules under test	1. Registration & Login 2. Vendor Profile & Daily Sales 3. Trust Scoring Engine 4. Loan Application & Bank Review 5. EMI Schedule & Repayment (Module 4 continued)
Test levels	System testing through the web interface: 80 cases - the tester fills forms and clicks buttons; the output is the message or value shown on screen. Unit testing of the Trust Score formula: 12 cases. Combination testing: 1,894 automatically generated combinations of valid and invalid field values in 7 forms (Section 8). Integration testing: 2 end-to-end scenarios (tests/test_integration.py).
Design techniques	Positive testing (valid input); equivalence partitioning - every invalid class of every field (empty, wrong format, non-numeric, nan / inf, out of range); boundary-value analysis on both sides of every limit (name 50 / 51, password 5 / 6, amount 1,00,000 / 1,00,000.01, date 90 / 91 days, score 39 / 40 and 69 / 70, tier limit 15,000 / 15,000.01, EMI 1 paisa short, 90 / 91 days overdue); combination testing (several invalid fields at once, decision tables); state-based testing (loan life cycle); security testing (role check, forged requests, SQL injection).
Test runs	Run 1 - the first version of the code (git commit 0ce1911): 9 test cases FAILED and 6 defects were logged (Section 9). Run 2 - after fixing the defects, the same cases were executed again (regression testing); the results in Sections 3-8 are from Run 2.
Environment	Windows 11, Python 3.11, Flask 3.1, SQLite 3; Chrome / Edge browser.
Tools	pytest; Flask test client (drives the real web routes); tests/run_test_cases.py records each Actual Output automatically.
Test data	Every case starts from a fresh, empty database; the data each case needs is listed in its Pre-condition column.
Entry criteria	Code of all four modules complete; application starts without errors.
Exit criteria	All documented test cases and combinations executed; every defect found fixed and re-tested; every Must-have requirement traced to at least one passing test case (Section 10).

2. Test Results Summary

Module	Test case IDs	Cases	Run 1 passed	Run 1 failed	Run 2 passed	Run 2 failed	Final pass rate
User Registration & Login	TC-AUTH-01 to TC-AUTH-20	20	18	2	20	0	100%
Vendor Profile & Daily Sales Logging	TC-SALE-01 to TC-SALE-19	19	17	2	19	0	100%
Trust Score Calculation	TC-TS-01 to TC-TS-14	14	14	0	14	0	100%
Loan Application & Bank Review	TC-LOAN-01 to TC-LOAN-21	21	19	2	21	0	100%
EMI Schedule & Repayment	TC-EMI-01 to TC-EMI-18	18	15	3	18	0	100%
Total		**92**	**83**	**9**	**92**	**0**	**100%**

Test cases by design technique (EP = equivalence partitioning, BVA = boundary-value analysis; a BVA case tests the values on both sides of a limit):

Module	Positive	Negative	Negative (EP)	BVA	Combination	Security	Total
User Registration & Login	2	4	6	4	1	3	20
Vendor Profile & Daily Sales Logging	2	2	7	6	1	1	19
Trust Score Calculation	5	3	0	6	0	0	14
Loan Application & Bank Review	3	7	3	3	1	4	21
EMI Schedule & Repayment	6	3	3	2	0	4	18

Module	Positive	Negative	Negative (EP)	BVA	Combination	Security	Total
Total	**18**	**19**	**19**	**21**	**3**	**12**	**92**

Combination testing: 1,894 of 1,894 combinations gave the expected result in Run 2 (Run 1: 1,567 of 1,894, with 9 server crashes) - see Section 8.

Automated run (pytest): 101 passed in 108.62s (0:01:48) - the 92 documented cases above, the 7 combination suites and 2 integration tests. Executed on 04-11-2026.

How the Actual Output is obtained: tests/run_test_cases.py performs each case's steps on the running application and copies the message shown on the page into the Actual Output column; the Remark is PASS only if it matches the Expected Output exactly. Nothing in the Actual Output column is typed by hand.

Repeating a case by hand in the demo: the demo data (python seed_demo.py) already contains loans, so loan numbers such as #1 will differ; the messages are otherwise identical.

3. User Registration & Login

Test Case #:	TC-AUTH-01 to TC-AUTH-20	**Test Case Name:**	User Registration & Login
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026
Short Description:	Verify account creation rules (valid / invalid classes, boundaries, several errors at once), login, lock-out after 5 wrong passwords and protection of pages that need a login.		

Pre-conditions
1. Application is running with an empty database.
2. User is on the Register or Log in page of VendorSetu.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-AUTH-01	Register with valid details	System Positive	No account exists for 9876543210	1. Open the Register page 2. Enter name, mobile, role, password and confirm password 3. Click 'Create account'	Name = Ramesh Kumar, Mobile = 9876543210, Role = Street Vendor, Password = ramesh123, Confirm = ramesh123	Registration successful. Please log in. -> Log in page is shown	Registration successful. Please log in. -> Log in page is shown	**PASS**
TC-AUTH-02	Register with an already registered mobile number	System Negative	An account already exists for 9876543210	1. Open the Register page 2. Enter the same mobile number with other details 3. Click 'Create account'	Name = Ramesh Kumar, Mobile = 9876543210 (already registered), Password = ramesh123	Mobile number already registered	Mobile number already registered	**PASS**
TC-AUTH-03	Register with an invalid mobile number	System Negative (EP)	-	1. Open the Register page 2. Enter a 5-digit mobile number 3. Click 'Create account'	Mobile = 98765 (5 digits), other fields valid	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**PASS**
TC-AUTH-04	Register with a weak password	System Negative (EP)	-	1. Open the Register page 2. Enter a 3-letter password in both password fields 3. Click 'Create account'	Password = abc, Confirm = abc, other fields valid	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	**PASS**
TC-AUTH-05	Register with non-matching passwords	System Negative	-	1. Open the Register page 2. Enter different values in Password and Confirm password 3. Click 'Create account'	Password = ramesh123, Confirm = ramesh124	Passwords do not match	Passwords do not match	**PASS**
TC-AUTH-06	Log in with correct credentials	System Positive	Ramesh is registered as a vendor	1. Open the Log in page 2. Enter mobile number and password 3. Click 'Log in'	Mobile = 9876543210, Password = ramesh123	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	**PASS**
TC-AUTH-07	Account lock after 5 wrong passwords	System BVA	Ramesh is registered as a vendor	1. Log in with a wrong password 5 times 2. Then log in with the correct password	Mobile = 9876543210; Password = wrong1 (x5), then ramesh123	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	**PASS**
TC-AUTH-08	Register with the name left empty	System Negative (EP)	-	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Name = (empty), other fields valid	Name is required	Name is required	**PASS**
TC-AUTH-09	Register with digits as the name	System Negative (EP)	-	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Name = 12345, other fields valid	Name can contain only letters, spaces and dots	Name can contain only letters, spaces and dots	**PASS** (Run 1: FAIL, DEF-04 fixed)

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-AUTH-10	Boundary value - name length 50 / 51	System BVA	No account exists for 9876543210	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Name = 'A' x 50 characters; then Name = 'A' x 51 characters (fresh database each time)	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	**PASS** (Run 1: FAIL, DEF-04 fixed)
TC-AUTH-11	Mobile number - invalid equivalence classes	System Negative (EP)	-	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Mobile = 987654321 (9 digits) / 98765432101 (11 digits) / 5876543210 (starts with 5) / 98765abcde	9 digits: Enter a valid 10-digit mobile number | 11 digits: Enter a valid 10-digit mobile number | Starts with 5: Enter a valid 10-digit mobile number | Letters: Enter a valid 10-digit mobile number	9 digits: Enter a valid 10-digit mobile number | 11 digits: Enter a valid 10-digit mobile number | Starts with 5: Enter a valid 10-digit mobile number | Letters: Enter a valid 10-digit mobile number	**PASS**
TC-AUTH-12	Boundary value - password length 5 / 6	System BVA	No account exists for 9876543210	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Password = ab123 (5 characters); then Password = ab1234 (6 characters)	5 characters: Password must be at least 6 characters with letters and digits | 6 characters: Registration successful. Please log in.	5 characters: Password must be at least 6 characters with letters and digits | 6 characters: Registration successful. Please log in.	**PASS**
TC-AUTH-13	Password with only letters / only digits	System Negative (EP)	-	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	Password = abcdefgh (no digit); then Password = 12345678 (no letter)	Only letters: Password must be at least 6 characters with letters and digits | Only digits: Password must be at least 6 characters with letters and digits	Only letters: Password must be at least 6 characters with letters and digits | Only digits: Password must be at least 6 characters with letters and digits	**PASS**
TC-AUTH-14	No role / forged role	System Security	-	1. Open the Register page 2. Do not select a role (or send role=ADMIN in the request) 3. Click 'Create account'	Role = (not selected); then Role = ADMIN (not in the list)	Not selected: Invalid role selected | ADMIN: Invalid role selected	Not selected: Invalid role selected | ADMIN: Invalid role selected	**PASS**
TC-AUTH-15	Several invalid fields at once (combination)	System Combination	-	1. Open the Register page 2. Fill the form with the input values 3. Click 'Create account'	(a) all fields empty; (b) Mobile = 98765 and Password = abc together	All empty: Name is required | Mobile + password invalid: Enter a valid 10-digit mobile number	All empty: Name is required | Mobile + password invalid: Enter a valid 10-digit mobile number	**PASS**
TC-AUTH-16	Log in with a wrong password	System Negative	Ramesh is registered as a vendor	1. Open the Log in page 2. Enter the mobile number and password 3. Click 'Log in'	Mobile = 9876543210, Password = ramesh124	Invalid mobile number or password	Invalid mobile number or password	**PASS**
TC-AUTH-17	Log in with an unregistered mobile number	System Negative	Ramesh is registered as a vendor	1. Open the Log in page 2. Enter the mobile number and password 3. Click 'Log in'	Mobile = 9000000001 (not registered), Password = ramesh123	Invalid mobile number or password	Invalid mobile number or password	**PASS**
TC-AUTH-18	SQL injection in the login form	System Security	Ramesh is registered as a vendor	1. Open the Log in page 2. Enter the mobile number and password 3. Click 'Log in'	Mobile = ' OR '1'='1, Password = ' OR '1'='1	Invalid mobile number or password	Invalid mobile number or password	**PASS**
TC-AUTH-19	Failed-attempt counter resets after a successful login	System BVA	Ramesh is registered as a vendor	1. Log in with a wrong password 4 times, then the correct one 2. Log in with a wrong password 4 more times, then the correct one	Mobile = 9876543210; wrong1 (x4), ramesh123, wrong1 (x4), ramesh123	4 wrong + correct: Welcome, Ramesh Kumar! | 4 more wrong: Invalid mobile number or password | Correct again: Welcome, Ramesh Kumar!	4 wrong + correct: Welcome, Ramesh Kumar! | 4 more wrong: Invalid mobile number or password | Correct again: Welcome, Ramesh Kumar!	**PASS**
TC-AUTH-20	Open the dashboard without logging in	System Security	Nobody is logged in	1. Type the dashboard address in the browser	URL = /dashboard	Please log in first. -> Log in page is shown	Please log in first. -> Log in page is shown	**PASS**

Post-conditions

1. A valid registration stores exactly one user with a hashed password; rejected input stores nothing.
2. After 5 consecutive wrong passwords the account stays locked even for the correct password.

4. Vendor Profile & Daily Sales Logging

Test Case #:	TC-SALE-01 to TC-SALE-19	**Test Case Name:**	Vendor Profile & Daily Sales Logging
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026
Short Description:	Verify the business profile rules and that a vendor can log one valid sales entry per day, while invalid amounts, dates and payment modes are rejected without saving anything.		

Pre-conditions
1. Vendor 'Ramesh Kumar' (9876543210 / ramesh123) is registered and logged in.
2. His business profile is saved (unless the case says otherwise).

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-SALE-01	Log a valid daily sale	System Positive	No sale logged today	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026 (today), Amount = 1500, Mode = UPI	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**PASS**
TC-SALE-02	Log a sale with zero amount	System BVA	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = 0, Mode = Cash	Sale amount must be greater than 0	Sale amount must be greater than 0	**PASS**
TC-SALE-03	Log a sale for a future date	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 06-11-2026 (2 days ahead), Amount = 1200, Mode = Cash	Sale date cannot be in the future	Sale date cannot be in the future	**PASS**
TC-SALE-04	Log a second sale for the same date	System Negative	A sale for 03-11-2026 is already logged	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 03-11-2026, Amount = 900, Mode = Cash	Sales already logged for this date	Sales already logged for this date	**PASS**
TC-SALE-05	Log a sale above the daily limit	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = 150000, Mode = UPI	Sale amount cannot exceed Rs 1,00,000 per day	Sale amount cannot exceed Rs 1,00,000 per day	**PASS**
TC-SALE-06	Log a sale before saving the business profile	System Negative	Ramesh has NOT saved his business profile	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = 1500, Mode = UPI	Complete your vendor profile first	Complete your vendor profile first	**PASS**
TC-SALE-07	30-day summary on the dashboard	System Positive	No sales logged yet	1. Log three sales (one per day) 2. Open the dashboard and read the summary cards	02-11-2026: 1000, 03-11-2026: 2000, 04-11-2026: 3000	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	**PASS**
TC-SALE-08	Log a sale with a negative amount	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = -500, Mode = Cash	Sale amount must be greater than 0	Sale amount must be greater than 0	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-SALE-09	Non-numeric / empty amount	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = abc; then Amount = (empty)	abc: Sale amount must be a number | Empty: Sale amount must be a number	abc: Sale amount must be a number | Empty: Sale amount must be a number	**PASS**
TC-SALE-10	Special float values as the amount	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = nan; then Amount = inf	nan: Sale amount must be a number | inf: Sale amount must be a number	nan: Sale amount must be a number | inf: Sale amount must be a number	**PASS** (Run 1: FAIL, DEF-01 fixed)
TC-SALE-11	Boundary value - upper limit Rs 1,00,000	System BVA	No sale logged today	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = 100000; then Amount = 100000.01	100000: Sale logged. Your trust score has been updated. | 100000.01: Sale amount cannot exceed Rs 1,00,000 per day	100000: Sale logged. Your trust score has been updated. | 100000.01: Sale amount cannot exceed Rs 1,00,000 per day	**PASS**
TC-SALE-12	Boundary value - lower limit	System BVA	No sale logged today	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026, Amount = 1; then Amount = -0.01	1: Sale logged. Your trust score has been updated. | -0.01: Sale amount must be greater than 0	1: Sale logged. Your trust score has been updated. | -0.01: Sale amount must be greater than 0	**PASS**
TC-SALE-13	Payment mode other than Cash / UPI (forged request)	System Security	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. Send the sales form with a payment mode that is not in the list	Date = 04-11-2026, Amount = 1500, Mode = CARD; then Mode = (empty)	CARD: Payment mode must be CASH or UPI | Empty: Payment mode must be CASH or UPI	CARD: Payment mode must be CASH or UPI | Empty: Payment mode must be CASH or UPI	**PASS**
TC-SALE-14	Invalid / empty date	System Negative (EP)	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 2026-02-30 (no such day); then Date = (empty); Amount = 1500, Mode = UPI	2026-02-30: Enter a valid date (YYYY-MM-DD) | Empty: Enter a valid date (YYYY-MM-DD)	2026-02-30: Enter a valid date (YYYY-MM-DD) | Empty: Enter a valid date (YYYY-MM-DD)	**PASS**
TC-SALE-15	Boundary value - back-dated entry 90 / 91 days old	System BVA	No sales logged	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 06-08-2026 (90 days ago); then Date = 05-08-2026 (91 days ago); Amount = 1500	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	**PASS** (Run 1: FAIL, DEF-02 fixed)
TC-SALE-16	Boundary value - today / tomorrow	System BVA	No sale logged today	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026 (today); then Date = 05-11-2026 (tomorrow); Amount = 1500	Today: Sale logged. Your trust score has been updated. | Tomorrow: Sale date cannot be in the future	Today: Sale logged. Your trust score has been updated. | Tomorrow: Sale date cannot be in the future	**PASS**
TC-SALE-17	Several invalid fields at once (combination)	System Combination	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	(a) Amount = 0, Date = tomorrow, Mode = CARD; (b) Amount = 1500, Date = tomorrow, Mode = CARD	Amount + date + mode invalid: Sale amount must be greater than 0 | Date + mode invalid: Sale date cannot be in the future	Amount + date + mode invalid: Sale amount must be greater than 0 | Date + mode invalid: Sale date cannot be in the future	**PASS**
TC-SALE-18	Boundary value - years of vending 60 / 61 / -1	System BVA	Ramesh is registered but has NOT saved his business profile	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Business profile', enter the values 3. Click 'Save profile'	Business = Fruits & vegetables, Zone = Sector 14 Market, Years = 60; then 61; then -1	60: Profile saved. | 61: Years of vending must be between 0 and 60 | -1: Years of vending must be between 0 and 60	60: Profile saved. | 61: Years of vending must be between 0 and 60 | -1: Years of vending must be between 0 and 60	**PASS**
TC-SALE-19	Profile with a mandatory field missing	System Negative (EP)	Ramesh is registered but has NOT saved his business profile	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Business profile', enter the values 3. Click 'Save profile'	Business type = (empty); then Vending zone = (empty); then Years = abc	No business type: Business type and vending zone are required | No zone: Business type and vending zone are required | Years = abc: Years of vending must be a number	No business type: Business type and vending zone are required | No zone: Business type and vending zone are required | Years = abc: Years of vending must be a number	**PASS**

Post-conditions
1. A valid entry is saved and counted in the 30-day summary; a rejected entry changes nothing. 2. A rejected profile is not saved.

5. Trust Score Calculation

Test Case #:	TC-TS-01 to TC-TS-14	**Test Case Name:**	Trust Score Calculation
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026
Short Description:	Verify the 0-100 Trust Score formula, the risk band and the loan-eligibility cut-off of 40.		

Pre-conditions
1. Formula: Sales (active days / 30 x 40) + Repayment (on-time / due x 40, or 20 if no EMI is due yet) + Peer vouching (min(vouches, 3) / 3 x 10) + Tenure (min(years, 5) / 5 x 10), rounded.
2. Bands: 70-100 LOW RISK, 40-69 MEDIUM RISK, 0-39 HIGH RISK.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-TS-01	Ideal vendor - maximum observable score	Unit Positive	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 30/30, EMIs due = 0, Vouches = 3, Years = 5	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	**PASS**
TC-TS-02	Boundary value - score exactly 40	Unit BVA	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 15/30, EMIs due = 0, Vouches = 0, Years = 0	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	**PASS**
TC-TS-03	Boundary value - score just below 40	Unit BVA	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 14/30, EMIs due = 0, Vouches = 0, Years = 0	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	**PASS**
TC-TS-04	Vendor with a partial repayment record	Unit Positive	-	1. Call compute_trust_score() with the input values	Active days = 24/30, On-time EMIs = 6 of 12 due, Vouches = 1, Years = 2	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	**PASS**
TC-TS-05	Invalid input - active days exceed the window	Unit Negative	-	1. Call compute_trust_score() with the input values	Active days = 35, Window = 30	Active days cannot exceed the observation window	Active days cannot exceed the observation window	**PASS**
TC-TS-06	New vendor sees the score on the dashboard	System Positive	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	1. Log in as Sunita 2. Read the Trust Score card	Mobile = 9812345670, Password = sunita123	Dashboard shows Trust Score 20/100, HIGH RISK	Dashboard shows Trust Score 20/100, HIGH RISK	**PASS**
TC-TS-07	Minimum score - brand-new vendor	Unit BVA	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 0/30, EMIs due = 0, Vouches = 0, Years = 0	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	0 + 20 + 0 + 0 = 20 -> HIGH RISK, not eligible	**PASS**
TC-TS-08	Maximum score - perfect record	Unit BVA	-	1. Call compute_trust_score() with the input values	Active days = 30/30, On-time EMIs = 12 of 12 due, Vouches = 3, Years = 5	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	40 + 40 + 10 + 10 = 100 -> LOW RISK, eligible	**PASS**
TC-TS-09	Vouches and years above their caps	Unit BVA	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 30/30, EMIs due = 0, Vouches = 7 (cap 3), Years = 20 (cap 5)	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	**PASS**
TC-TS-10	Good sales but no EMI paid on time	Unit Positive	-	1. Call compute_trust_score() with the input values	Active days = 30/30, On-time EMIs = 0 of 6 due, Vouches = 0, Years = 5	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	40 + 0 + 0 + 10 = 50 -> MEDIUM RISK, eligible	**PASS**
TC-TS-11	Boundary value - score 69 / 70 (MEDIUM / LOW RISK)	Unit BVA	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 29/30, then 30/30; EMIs due = 0, Vouches = 0, Years = 5	29 days: 38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible | 30 days: 40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	29 days: 38.7 + 20 + 0 + 10 = 69 -> MEDIUM RISK, eligible | 30 days: 40 + 20 + 0 + 10 = 70 -> LOW RISK, eligible	**PASS**
TC-TS-12	Invalid input - negative vouches	Unit Negative	-	1. Call compute_trust_score() with the input values	Active days = 30/30, EMIs due = 0, Vouches = -1, Years = 5	Score inputs cannot be negative	Score inputs cannot be negative	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-TS-13	Invalid input - on-time EMIs exceed EMIs due	Unit Negative	-	1. Call compute_trust_score() with the input values	Active days = 30/30, On-time EMIs = 5, EMIs due = 3, Vouches = 0, Years = 5	On-time EMIs cannot exceed EMIs due	On-time EMIs cannot exceed EMIs due	**PASS**
TC-TS-14	Score goes up after a sale is logged	System Positive	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	1. Log in as Sunita and read the score 2. Log today's sale 3. Read the score again	Date = 04-11-2026, Amount = 800, Mode = Cash	Before: 20/100 | After logging 1 sale: 21/100, HIGH RISK	Before: 20/100 | After logging 1 sale: 21/100, HIGH RISK	**PASS**

Post-conditions

1. Score, band and eligibility are returned / displayed; no stored data is modified.

6. Loan Application & Bank Review

Test Case #:	TC-LOAN-01 to TC-LOAN-21	**Test Case Name:**	Loan Application & Bank Review
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026
Short Description:	Verify progressive-tier loan application rules and the bank officer's approve / reject / disburse actions.		

Pre-conditions
1. Vendor 'Ramesh Kumar' (9876543210 / ramesh123): profile with 5 years of vending and a sale logged on each of the last 30 days, so his Trust Score is 40 + 20 + 0 + 10 = 70.
2. Bank officer 'Anil Sharma' (9123456780 / bank123) is registered.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-LOAN-01	Eligible vendor applies for a Tier-1 loan	System BVA	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = 15000 (Trust Score 70, Tier 1 limit 15,000)	Loan application #1 submitted. Status: PENDING.	Loan application #1 submitted. Status: PENDING.	**PASS**
TC-LOAN-02	Amount above the tier limit	System Negative (EP)	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. Enter an amount above Rs 15,000 3. Click 'Apply for loan'	Amount = 20000	Amount exceeds Tier 1 limit of Rs 15,000	Amount exceeds Tier 1 limit of Rs 15,000	**PASS**
TC-LOAN-03	Vendor with a low Trust Score applies	System Negative	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales (score 20)	1. Log in as Sunita 2. Enter the amount 3. Click 'Apply for loan'	Amount = 10000	Trust score 20 is below 40. Not eligible	Trust score 20 is below 40. Not eligible	**PASS**
TC-LOAN-04	Second application while one is pending	System Negative	Ramesh's application #1 for Rs 15,000 is PENDING	1. Log in as Ramesh (9876543210 / ramesh123) 2. Enter the amount 3. Click 'Apply for loan'	Amount = 5000	You already have an active loan	You already have an active loan	**PASS**
TC-LOAN-05	Officer rejects without remarks	System Negative	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Leave the Remarks box empty 3. Click 'Reject' on application #1	Remarks = (empty), Decision = Reject	Remarks are required when rejecting	Remarks are required when rejecting	**PASS**
TC-LOAN-06	Vendor tries to open the bank officer page	System Security	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. Type the officer page address in the browser	URL = /officer	You are not allowed to open that page.	You are not allowed to open that page.	**PASS**
TC-LOAN-07	Officer approves and disburses	System Positive	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Click 'Approve' on application #1 3. Click 'Disburse' on application #1	Decision = Approve, then Disburse	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	**PASS**
TC-LOAN-08	Zero / negative loan amount	System BVA	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = 0; then Amount = -5000	0: Loan amount must be greater than 0 | -5000: Loan amount must be greater than 0	0: Loan amount must be greater than 0 | -5000: Loan amount must be greater than 0	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-LOAN-09	Non-numeric / empty loan amount	System Negative (EP)	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = abc; then Amount = (empty)	abc: Loan amount must be a number | Empty: Loan amount must be a number	abc: Loan amount must be a number | Empty: Loan amount must be a number	**PASS**
TC-LOAN-10	Special float values as the loan amount	System Negative (EP)	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = nan; then Amount = inf	nan: Loan amount must be a number | inf: Loan amount must be a number	nan: Loan amount must be a number | inf: Loan amount must be a number	**PASS** (Run 1: FAIL, DEF-01 fixed)
TC-LOAN-11	Boundary value - Tier-1 limit 15,000 / 15,000.01	System BVA	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = 15000; then Amount = 15000.01 (fresh database each time)	15000: Loan application #1 submitted. Status: PENDING. | 15000.01: Amount exceeds Tier 1 limit of Rs 15,000	15000: Loan application #1 submitted. Status: PENDING. | 15000.01: Amount exceeds Tier 1 limit of Rs 15,000	**PASS**
TC-LOAN-12	Apply before saving the business profile	System Negative	Ramesh is registered but has NOT saved his business profile	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = 10000	Complete your vendor profile first	Complete your vendor profile first	**PASS**
TC-LOAN-13	Two rules broken at once (combination)	System Combination	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	(a) Sunita, score 20, Amount = 20000 (low score + above limit); (b) Ramesh with application #1 PENDING, Amount = 20000 (active loan + above limit)	Low score + above limit: Trust score 20 is below 40. Not eligible | Active loan + above limit: You already have an active loan	Low score + above limit: Trust score 20 is below 40. Not eligible | Active loan + above limit: You already have an active loan	**PASS**
TC-LOAN-14	Review an application that is already approved	System Negative	Ramesh's application #1 is APPROVED	1. Log in as bank officer (9123456780 / bank123) 2. Send a Reject request for application #1	Decision = Reject, Remarks = Changed my mind	Only PENDING applications can be reviewed	Only PENDING applications can be reviewed	**PASS**
TC-LOAN-15	Disburse an application that is still pending	System Negative	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Send a Disburse request for application #1	POST /officer/loan/1/disburse	Only APPROVED loans can be disbursed	Only APPROVED loans can be disbursed	**PASS**
TC-LOAN-16	Forged decision value	System Security	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Send the review form with a decision that is not offered	Decision = DELETE, Remarks = x	Decision must be APPROVE or REJECT	Decision must be APPROVE or REJECT	**PASS**
TC-LOAN-17	Review a loan that does not exist	System Negative	Only application #1 exists	1. Log in as bank officer (9123456780 / bank123) 2. Send an Approve request for application #99	POST /officer/loan/99/review, Approve	Loan not found	Loan not found	**PASS**
TC-LOAN-18	Withdraw a pending application, then apply again	System Positive	Ramesh's application #1 for Rs 15,000 is PENDING	1. Log in as Ramesh (9876543210 / ramesh123) 2. Click 'Withdraw' on application #1 3. Apply for Rs 12,000	Withdraw #1; then Amount = 12000	Application #1 withdrawn. | Loan application #2 submitted. Status: PENDING.	Application #1 withdrawn. | Loan application #2 submitted. Status: PENDING.	**PASS**
TC-LOAN-19	Reject with remarks; vendor sees them and can re-apply	System Positive	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Enter remarks and click 'Reject' on #1 3. Log in as Ramesh (9876543210 / ramesh123) 4. Open loan #1 5. Apply for Rs 15,000 again	Remarks = Sales record too short; then Amount = 15000	Loan #1 REJECTED. | Vendor's loan page shows remarks: Sales record too short | Loan application #2 submitted. Status: PENDING.	Loan #1 REJECTED. | Vendor's loan page shows remarks: Sales record too short | Loan application #2 submitted. Status: PENDING.	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-LOAN-20	Customer / municipal officer opens a vendor's loan page	System Security	Ramesh's application #1 exists	1. Log in as customer Priya (9988776655 / priya123) or municipal officer (9090909090 / muni123) 2. Type /loan/1 in the browser	URL = /loan/1	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	**PASS** (Run 1: FAIL, DEF-05 fixed)
TC-LOAN-21	Vendor opens another vendor's loan page	System Security	Ramesh's application #1 exists	1. Log in as Sunita (9812345670 / sunita123) 2. Type /loan/1 in the browser	URL = /loan/1	You can only view your own loans.	You can only view your own loans.	**PASS**

Post-conditions

1. An accepted application is stored as PENDING; approval and disbursal change its status and create the EMI schedule; rejected requests change nothing.
2. Only the owning vendor and bank officers can open a loan's page.

7. EMI Schedule & Repayment

Test Case #:	TC-EMI-01 to TC-EMI-18	**Test Case Name:**	EMI Schedule & Repayment
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	04-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	04-11-2026
Short Description:	Verify the EMI schedule, the repayment rules, loan closure (next tier unlocked) and the default check.		

Pre-conditions
1. Ramesh's Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months has been approved and disbursed today by the bank officer (unless the case says otherwise).
2. Hand calculation: r = 12% / 12 = 0.01, EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-EMI-01	View the EMI schedule	System Positive	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. Click 'View' on loan #1 3. Read the EMI schedule table	Loan #1: P = 15000, r = 12% p.a., n = 12, disbursed 04-11-2026	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	12 installments; EMI #1 Rs 1,332.73 due 2026-12-04; EMI #12 due 2027-11-04	**PASS**
TC-EMI-02	Pay EMI #1 before its due date	System Positive	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Click 'Pay' on installment #1 (amount pre-filled)	Installment = 1, Amount = 1332.73, paid on 04-11-2026	EMI #1 paid on time.	EMI #1 paid on time.	**PASS**
TC-EMI-03	Pay less than the EMI amount	System Negative	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Change the amount to 1000 4. Click 'Pay' on installment #1	Installment = 1, Amount = 1000	Amount is less than EMI due of Rs 1332.73	Amount is less than EMI due of Rs 1332.73	**PASS**
TC-EMI-04	Skip an installment (forged request)	System Security	EMI #1 is paid; #2 is due	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay EMI #1 3. Send a payment request for installment #3 directly (the page only offers #2)	POST /vendor/loan/1/pay/3, Amount = 1332.73	Pay earlier installment #2 first	Pay earlier installment #2 first	**PASS**
TC-EMI-05	Pay the same installment twice (forged request)	System Security	EMI #1 is already paid	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay EMI #1 3. Send a payment request for installment #1 again	POST /vendor/loan/1/pay/1, Amount = 1332.73	Installment already paid	Installment already paid	**PASS**
TC-EMI-06	Repay all 12 EMIs (progressive lending)	System Positive	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay installments 1 to 12 in order 3. Open the dashboard	12 payments, each equal to the EMI shown	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	**PASS**
TC-EMI-07	Default check marks a 90+ days overdue loan	System Positive	Loan #1 was disbursed on 07-06-2026 (150 days ago) and no EMI has been paid	1. Log in as bank officer (9123456780 / bank123) 2. Click 'Run default check (90+ days overdue)'	(button click, no data)	Default check complete: 1 loan(s) marked DEFAULTED.	Default check complete: 1 loan(s) marked DEFAULTED.	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-EMI-08	Special float values as the payment	System Negative (EP)	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay'	Installment = 1, Amount = nan; then Amount = inf	nan: Payment amount must be a number | inf: Payment amount must be a number	nan: Payment amount must be a number | inf: Payment amount must be a number	**PASS** (Run 1: FAIL, DEF-03 fixed)
TC-EMI-09	Non-numeric / empty payment	System Negative (EP)	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay'	Installment = 1, Amount = abc; then Amount = (empty)	abc: Payment amount must be a number | Empty: Payment amount must be a number	abc: Payment amount must be a number | Empty: Payment amount must be a number	**PASS**
TC-EMI-10	Boundary value - 1 paisa less than the EMI / exact EMI	System BVA	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay'	Installment = 1, Amount = 1332.72; then Amount = 1332.73 (fresh database each time)	1332.72: Amount is less than EMI due of Rs 1332.73 | 1332.73: EMI #1 paid on time.	1332.72: Amount is less than EMI due of Rs 1332.73 | 1332.73: EMI #1 paid on time.	**PASS**
TC-EMI-11	Zero / negative payment	System Negative (EP)	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay'	Installment = 1, Amount = 0; then Amount = -1332.73	0: Amount is less than EMI due of Rs 1332.73 | -1332.73: Amount is less than EMI due of Rs 1332.73	0: Amount is less than EMI due of Rs 1332.73 | -1332.73: Amount is less than EMI due of Rs 1332.73	**PASS**
TC-EMI-12	Pay an EMI after its due date	System Positive	Loan #1 was disbursed on 25-09-2026 (40 days ago), so EMI #1 is overdue	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay' 5. Open the dashboard	Installment = 1, Amount = 1332.73, paid on 04-11-2026	EMI #1 paid late. -> Trust Score 50/100	EMI #1 paid late. -> Trust Score 50/100	**PASS**
TC-EMI-13	Paying on time raises the Trust Score	System Positive	No EMI paid yet; Trust Score 70	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Enter the amount for the installment 4. Click 'Pay' 5. Open the dashboard	Installment = 1, Amount = 1332.73	Before: 70/100 | After paying EMI #1 on time: 90/100	Before: 70/100 | After paying EMI #1 on time: 90/100	**PASS**
TC-EMI-14	Installment number that does not exist (forged request)	System Security	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Send a payment request for installment #13, then #0	POST /vendor/loan/1/pay/13 and /pay/0, Amount = 1332.73	#13: Installment not found | #0: Installment not found	#13: Installment not found | #0: Installment not found	**PASS**
TC-EMI-15	Vendor pays another vendor's EMI (forged request)	System Security	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	1. Log in as Sunita (9812345670 / sunita123) 2. Send a payment request for Ramesh's loan #1	POST /vendor/loan/1/pay/1, Amount = 1332.73	You can only pay your own loans	You can only pay your own loans	**PASS** (Run 1: FAIL, DEF-06 fixed)
TC-EMI-16	Pay an EMI of a loan that does not exist	System Negative	Only loan #1 exists	1. Log in as Ramesh (9876543210 / ramesh123) 2. Send a payment request for loan #99	POST /vendor/loan/99/pay/1, Amount = 1332.73	Loan not found	Loan not found	**PASS** (Run 1: FAIL, DEF-06 fixed)
TC-EMI-17	Pay an EMI of an approved but not disbursed loan	System Negative	Ramesh's loan #1 is APPROVED (not disbursed)	1. Log in as Ramesh (9876543210 / ramesh123) 2. Send a payment request for loan #1	POST /vendor/loan/1/pay/1, Amount = 1332.73	Loan is not active for repayment	Loan is not active for repayment	**PASS**

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-EMI-18	Boundary value - EMI overdue 90 / 91 days	System BVA	Loan #1 is DISBURSED, no EMI paid	1. Log in as bank officer (9123456780 / bank123) 2. Click 'Run default check (90+ days overdue)'	EMI #1 due date set to 90 days ago; then to 91 days ago (fresh database each time)	90 days: Default check complete: 0 loan(s) marked DEFAULTED. | 91 days: Default check complete: 1 loan(s) marked DEFAULTED.	90 days: Default check complete: 0 loan(s) marked DEFAULTED. | 91 days: Default check complete: 1 loan(s) marked DEFAULTED.	**PASS**

Post-conditions

1. Paid installments are marked PAID (on time / late); when all 12 are paid the loan is CLOSED and Tier 2 (Rs 25,000) is unlocked.

8. Combination Testing

Single test cases check one rule at a time. To check that the rules also work together, every form was tested with **every combination** of the values below (cartesian product): valid values, every invalid class and the boundary values of each field. For each combination an independent oracle written from the SRS rules (tests/combinations.py) gives the message that must appear. A combination passes only if the message is exactly that, the app does not crash, and a rejected input leaves the database unchanged.

Form	Values tried for each field	Combinations	Run 1 (first version)	Run 2 (after fixes)	Result
Register	**Name:** 'Ramesh Kumar', 'Mohd. Salim', '', '12345', 'A' x 51 **Mobile:** '9876543210', '987654321', '98765432101', '5876543210', '98765abcde', '' **Password:** 'ramesh123', 'ab123', 'abcdefgh', '12345678', '' **Confirm:** 'same', 'different' **Role:** 'VENDOR', 'BANK_OFFICER', '', 'ADMIN'	1,200	960 matched, 240 wrong, 0 crashed	1,200 matched, 0 wrong, 0 crashed	PASS
Log in	**Mobile:** '9876543210', ' 9876543210 ', '9000000001', '', "' OR '1'='1" **Password:** 'ramesh123', 'RAMESH123', 'ramesh124', '', "' OR '1'='1"	25	25 matched, 0 wrong, 0 crashed	25 matched, 0 wrong, 0 crashed	PASS
Business profile	**Business:** 'Fruits', '', ' ' **Zone:** 'Sector 14', '' **Years:** '0', '5', '60', '61', '-1', 'abc', '', '2.5'	48	48 matched, 0 wrong, 0 crashed	48 matched, 0 wrong, 0 crashed	PASS
Daily sales	**Date:** '2026-11-04', '2026-11-03', '2026-08-06', '2026-08-05', '2026-11-05', '2026-02-30', '', 'abc' **Amount:** '1500', '1', ' 250 ', '1e3', '100000', '100000.01', '0', '-5', 'abc', '', 'nan', 'inf' **Mode:** 'UPI', 'CASH', 'upi', 'CARD', ''	480	405 matched, 75 wrong, 8 crashed	480 matched, 0 wrong, 0 crashed	PASS
Loan application	**Vendor:** 'eligible (score 70)', 'score 20', 'no profile', 'has a pending loan' **Amount:** '15000', '1', '15000.01', '0', '-1', 'abc', '', 'nan', 'inf'	36	28 matched, 8 wrong, 1 crashed	36 matched, 0 wrong, 0 crashed	PASS
EMI payment	**Installment:** 1, 2, 12, 13, 0 **Amount:** '1332.73', '1332.72', '5000', '0', '-1', 'abc', '', 'nan', 'inf'	45	43 matched, 2 wrong, 0 crashed	45 matched, 0 wrong, 0 crashed	PASS
Role x page access	**Role:** 'Guest (not logged in)', 'Owner vendor (Ramesh)', 'Other vendor (Sunita)', 'Bank officer', 'Customer', 'Municipal officer' **Request:** 'GET /dashboard', 'GET /officer', 'GET /loan/1', 'POST /vendor/sales', 'POST /vendor/profile', 'POST /vendor/loan/apply', 'POST /vendor/loan/1/pay/1', 'POST /officer/loan/1/review', 'POST /officer/loan/1/disburse', 'POST /officer/mark-defaults'	60	58 matched, 2 wrong, 0 crashed	60 matched, 0 wrong, 0 crashed	PASS
Total		**1,894**	**1,567 matched**	**1,894 matched**	

Rule order when several fields are wrong: the first broken rule in form order is reported - Register: passwords match, name, mobile, password, role; Daily sales: date, amount, date range, payment mode, profile, duplicate date; Loan: amount format, active loan, Trust Score, amount range. TC-AUTH-15, TC-SALE-17 and TC-LOAN-13 are documented examples.

Role x page access: each of the 6 roles sends each of the 10 requests. Expected: a guest is sent to Log in; officer pages only for the bank officer; a loan page only for its vendor and bank officers (another vendor sees 'own loans only'); vendor forms only for vendors.

9. Defect Report (Run 1) and Re-test (Run 2)

Run 1 executed the same test cases on the first version of the code (git commit 0ce1911). 9 cases FAILED; their real Actual Output is shown below. Each failure was logged as a defect and fixed, and the whole suite was executed again (regression testing).

TC ID	Test Scenario	Input	Expected Output	Actual Output (Run 1)	Remark	Actual Output (Run 2, after fix)	Remark
TC-AUTH-09	Register with digits as the name	Name = 12345, other fields valid	Name can contain only letters, spaces and dots	Registration successful. Please log in.	**FAIL** DEF-04	Name can contain only letters, spaces and dots	**PASS**
TC-AUTH-10	Boundary value - name length 50 / 51	Name = 'A' x 50 characters; then Name = 'A' x 51 characters (fresh database each time)	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	50 characters: Registration successful. Please log in. | 51 characters: Registration successful. Please log in.	**FAIL** DEF-04	50 characters: Registration successful. Please log in. | 51 characters: Name must be at most 50 characters	**PASS**
TC-SALE-10	Special float values as the amount	Date = 04-11-2026, Amount = nan; then Amount = inf	nan: Sale amount must be a number | inf: Sale amount must be a number	nan: HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: sales.amount) | inf: Sale amount cannot exceed Rs 1,00,000 per day	**FAIL** DEF-01	nan: Sale amount must be a number | inf: Sale amount must be a number	**PASS**
TC-SALE-15	Boundary value - back-dated entry 90 / 91 days old	Date = 06-08-2026 (90 days ago); then Date = 05-08-2026 (91 days ago); Amount = 1500	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale logged. Your trust score has been updated.	**FAIL** DEF-02	90 days old: Sale logged. Your trust score has been updated. | 91 days old: Sale date cannot be more than 90 days old	**PASS**
TC-LOAN-10	Special float values as the loan amount	Amount = nan; then Amount = inf	nan: Loan amount must be a number | inf: Loan amount must be a number	nan: HTTP 500 Internal Server Error (IntegrityError: NOT NULL constraint failed: loans.amount) | inf: Amount exceeds Tier 1 limit of Rs 15,000	**FAIL** DEF-01	nan: Loan amount must be a number | inf: Loan amount must be a number	**PASS**
TC-LOAN-20	Customer / municipal officer opens a vendor's loan page	URL = /loan/1	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	Customer: (no message) [POST-CONDITION FAILED: Ramesh's loan #1 page is shown] | Municipal officer: (no message) [POST-CONDITION FAILED: Ramesh's loan #1 page is shown]	**FAIL** DEF-05	Customer: You are not allowed to open that page. | Municipal officer: You are not allowed to open that page.	**PASS**
TC-EMI-08	Special float values as the payment	Installment = 1, Amount = nan; then Amount = inf	nan: Payment amount must be a number | inf: Payment amount must be a number	nan: EMI #1 paid on time. | inf: EMI #1 paid on time.	**FAIL** DEF-03	nan: Payment amount must be a number | inf: Payment amount must be a number	**PASS**
TC-EMI-15	Vendor pays another vendor's EMI (forged request)	POST /vendor/loan/1/pay/1, Amount = 1332.73	You can only pay your own loans	You can only pay your own loans | You can only view your own loans.	**FAIL** DEF-06	You can only pay your own loans	**PASS**
TC-EMI-16	Pay an EMI of a loan that does not exist	POST /vendor/loan/99/pay/1, Amount = 1332.73	Loan not found	Loan not found | Loan not found	**FAIL** DEF-06	Loan not found	**PASS**

Defect log

Defect ID	Severity	Found by	What happened (Run 1)	Cause	Fix	Re-test (Run 2)
DEF-01	High	TC-SALE-10, TC-LOAN-10	Typing 'nan' as a sale or loan amount crashed the app (HTTP 500 Internal Server Error); 'inf' as a loan amount gave a misleading message.	Python's float() accepts 'nan' and 'inf'. NaN passed every comparison and SQLite stored it as NULL in a NOT NULL column.	New helper parse_number() in database.py rejects values that are not finite numbers.	PASS - closed
DEF-02	Medium	TC-SALE-15	A sale dated 1900-01-01 (or any old date) was accepted.	Only future dates were checked.	vendor.log_daily_sale() rejects entries more than 90 days old (MAX_BACKDATE_DAYS); FR-VEN-5 updated.	PASS - closed
DEF-03	High	TC-EMI-08	Paying an EMI with amount 'nan' or 'inf' marked the installment PAID.	Same as DEF-01: the 'less than EMI' check is False for NaN.	loan.pay_emi() uses parse_number().	PASS - closed
DEF-04	Low	TC-AUTH-09, TC-AUTH-10	Names such as '12345' or a 300-character string were accepted at registration.	The name was only checked for being non-empty.	auth.validate_name(): letters, spaces, dots and apostrophes only, at most 50 characters; FR-AUTH-1 updated.	PASS - closed

Defect ID	Severity	Found by	What happened (Run 1)	Cause	Fix	Re-test (Run 2)
DEF-05	High	TC-LOAN-20	A customer or municipal officer could open any vendor's loan page and see the amount and EMIs.	The /loan/<id> page only required a logged-in user and checked ownership for vendors only.	Only the owning vendor and bank officers may open a loan page (LOAN_VIEW_ROLES in app.py).	PASS - closed
DEF-06	Low	TC-EMI-15, TC-EMI-16	Paying an EMI of a loan that does not exist, or of another vendor's loan, showed two error messages ('Loan not found | Loan not found', 'You can only pay your own loans | You can only view your own loans.').	The error page redirected to the loan page, which reported the same error again.	The pay_emi route redirects to the dashboard when the loan is missing or not the vendor's own.	PASS - closed

10. Requirements Traceability Matrix

Each requirement from the SRS that the implemented modules cover, and the test cases that verify it.

Requirement	Verified by test case(s)	Status
FR-AUTH-1	TC-AUTH-01, TC-AUTH-05, TC-AUTH-08, TC-AUTH-09, TC-AUTH-10, TC-AUTH-14, TC-AUTH-15	Covered - PASS
FR-AUTH-2	TC-AUTH-02, TC-AUTH-03, TC-AUTH-11, TC-AUTH-15	Covered - PASS
FR-AUTH-3	TC-AUTH-04, TC-AUTH-12, TC-AUTH-13, TC-AUTH-15	Covered - PASS
FR-AUTH-5	TC-AUTH-06, TC-AUTH-16, TC-AUTH-17, TC-AUTH-18	Covered - PASS
FR-AUTH-6	TC-AUTH-07, TC-AUTH-19	Covered - PASS
FR-AUTH-7	TC-AUTH-20, TC-LOAN-06, TC-LOAN-20, TC-LOAN-21, TC-EMI-15	Covered - PASS
FR-VEN-2	TC-SALE-18, TC-SALE-19	Covered - PASS
FR-VEN-3	TC-SALE-01, TC-SALE-13, TC-SALE-14, TC-TS-14	Covered - PASS
FR-VEN-4	TC-SALE-02, TC-SALE-05, TC-SALE-08, TC-SALE-09, TC-SALE-10, TC-SALE-11, TC-SALE-12, TC-SALE-17	Covered - PASS
FR-VEN-5	TC-SALE-03, TC-SALE-04, TC-SALE-15, TC-SALE-16, TC-SALE-17	Covered - PASS
FR-VEN-6	TC-SALE-06, TC-LOAN-12	Covered - PASS
FR-VEN-7	TC-SALE-07	Covered - PASS
FR-TS-1	TC-TS-05, TC-TS-12, TC-TS-13, TC-TS-14, TC-LOAN-12	Covered - PASS
FR-TS-2	TC-TS-01, TC-TS-02, TC-TS-03, TC-TS-04, TC-TS-07, TC-TS-08, TC-TS-09, TC-TS-10, TC-EMI-12, TC-EMI-13	Covered - PASS
FR-TS-3	TC-TS-01, TC-TS-06, TC-TS-07	Covered - PASS
FR-TS-4	TC-TS-01, TC-TS-02, TC-TS-03, TC-TS-04, TC-TS-06, TC-TS-08, TC-TS-10, TC-TS-11	Covered - PASS
FR-TS-5	TC-TS-02, TC-TS-03, TC-LOAN-03, TC-LOAN-13	Covered - PASS
FR-TS-6	TC-TS-06	Covered - PASS
FR-LN-1	TC-LOAN-01	Covered - PASS
FR-LN-2	TC-LOAN-02, TC-LOAN-11	Covered - PASS
FR-LN-3	TC-EMI-06	Covered - PASS
FR-LN-4	TC-LOAN-02, TC-LOAN-08, TC-LOAN-09, TC-LOAN-10, TC-LOAN-11, TC-LOAN-13	Covered - PASS
FR-LN-5	TC-LOAN-04, TC-LOAN-13, TC-LOAN-18	Covered - PASS
FR-LN-6	TC-LOAN-01, TC-LOAN-18	Covered - PASS
FR-BNK-2	TC-LOAN-07, TC-LOAN-14, TC-LOAN-16, TC-LOAN-17	Covered - PASS
FR-BNK-3	TC-LOAN-05, TC-LOAN-19	Covered - PASS
FR-BNK-4	TC-LOAN-06	Covered - PASS
FR-BNK-5	TC-LOAN-07, TC-LOAN-15, TC-EMI-17	Covered - PASS
FR-BNK-6	TC-LOAN-07, TC-EMI-01	Covered - PASS
FR-BNK-7	TC-EMI-07, TC-EMI-18	Covered - PASS
FR-EMI-1	TC-EMI-01, TC-EMI-14, TC-EMI-16, TC-EMI-17	Covered - PASS
FR-EMI-2	TC-EMI-04	Covered - PASS
FR-EMI-3	TC-EMI-03, TC-EMI-08, TC-EMI-09, TC-EMI-10, TC-EMI-11	Covered - PASS
FR-EMI-4	TC-EMI-05	Covered - PASS
FR-EMI-5	TC-EMI-02, TC-EMI-12, TC-EMI-13	Covered - PASS
FR-EMI-6	TC-EMI-06	Covered - PASS
BR-1	TC-LOAN-06	Covered - PASS
BR-2	TC-LOAN-04	Covered - PASS
BR-4	TC-LOAN-03	Covered - PASS

Requirement	Verified by test case(s)	Status
BR-5	TC-LOAN-05, TC-LOAN-19	Covered - PASS
BR-6	TC-EMI-07, TC-EMI-18	Covered - PASS
NFR-SEC-2	TC-AUTH-07	Covered - PASS
NFR-SEC-4	TC-AUTH-18	Covered - PASS
NFR-SEC-7	TC-LOAN-20, TC-LOAN-21, TC-EMI-15	Covered - PASS

11. How to Re-execute the Tests

```
cd SE_Project
python tests/run_test_cases.py                (Run 2: every case + combination testing, PASS/FAIL)
python tests/run_test_cases.py --module LOAN  (one module only: AUTH, SALE, TS, LOAN or EMI)
python tests/run_test_cases.py --details      (also prints Input / Expected / Actual of each case)
python tests/run_test_cases.py --before-fix   (Run 1: same cases on the first version - shows FAILs)
python -m pytest tests -v                     (same cases, combination suites, integration tests)
python docs/build_test_cases.py               (regenerates this document from the latest runs)
```